M4GB divisibility signatures in update() (2026-11-12)

M4GB divisibility signatures in `update()` (2026-11-12)

Owner's ask (2026-11-12): "Speed it all up", after the Gröbner performance levers note (`docs/ic/GROEBNER_PERF_LEVERS_20261111.md`) and the persistent-server study (`../m4gb_persistent_server_20261111/`). This study measures the M4GB side. The in-house engines are a separate change.

Where the time went (Callgrind, before the change)

Captured 24-variable descent system (unsatisfiable, the common case), one thread, patches 0001–0003, `-O3 -march=x86-64-v3` (Valgrind cannot decode AVX-512), 2.15 billion instructions `[pcc:unverified]`:

part	share
`update()` (Gebauer–Möller pair update), inclusive	70 %
`divides` on integer-coded monomials	25 %
decoding an integer monomial one variable at a time (`const_iterator::operator++`)	16 %
GF(2) row additions (`add_to`)	3.4 %

M4GB stores monomials as their rank in degrevlex order. Every divisibility test against a leading monomial or a pair's lcm decodes that rank variable by variable, and `update()` runs such tests over every critical pair and every basis element for every new basis element. Bit-packing the GF(2) rows, lever 4 of the note, would touch 3.4 % of this profile; it is not the lever here.

What changed

Patch `0004-divisibility-signatures-in-update.patch` (switch `-DM4GB_SUPPORT_MASK=0` restores 0001–0003):

1. **Signature of a monomial**: `m1` has bit `v` set when variable `v` occurs, `m2` when it occurs with exponent 2; `small` says every exponent is at most 2 and `N <= 64`. Over GF(2) with the field equations every monomial is small.
2. **Exact divisibility from signatures**: `a | b` is `(a.m1 & ~b.m1) == 0 && (a.m2 & ~b.m2) == 0` when both are small (exponents in {0, 1, 2}); otherwise `m1` is only a necessary condition (valid for any `N`, bits fold modulo 64) and the exponent walk decides.
3. **Signatures of integer monomials from a cache**: a thread-local direct-mapped table of 16,384 entries keyed by the rank.
4. **lcm from signatures**: `set_lcm` writes the union of the masks when both operands are small, without decoding either.
5. **In `update(p)`**: the new leading monomial's signature is computed once; each critical pair stores its lcm's signature at construction; the pair pruning test, the "lcm(pLM, p_i) != lcm" tests and the minimal-lcm scan use the stored signatures.

Every result is decided exactly; nothing is approximated. Instruction counts on the two captured 24-variable systems (Callgrind, same build flags) `[pcc:unverified]`:

step	unsatisfiable system	satisfiable system
0001–0003	2.15 G	3.26 G
+ `m1` prefilter, cached	1.62 G	n/m
+ exact two-mask test	1.41 G	2.03 G
+ lcm from masks, signature of pLM once	1.28 G	1.80 G
+ plain thread-local table	1.24 G	1.76 G
+ signatures stored in critical pairs (0004)	1.01 G (0.47×)	1.39 G (0.43×)

After 0004 no function is above 7.5 % of the profile (`to_index` 7.3 %, `update` 6.6 %, `add_to` 6.4 %, map iteration 5.1 %).

Correctness before the run: on all 15 GF(2), GF(31) and GF(65521) inputs at hand (captured descent systems and the kernel study's inputs) the patched build and the switched-off build give the same exit code and byte-identical output files; the one 30-variable input timed out in both.

Protocol

1. `capture.sh` (with the 0001–0003 binaries): `ic descent --solver m4gb-f2 --solver fes-f2` on every target of 15 cells (Koblitz K and random R; m = 2 at 12 to 24 unknowns, m = 3 at 12 unknowns), 6 seeded targets each, through wrapper binaries that copy every input file: 90 systems.
2. `ab_corpus.py` under `tools/benchrec.py run`: three arms through `--serve` (one process per arm and binary width, warmed): `old` (0001–0003), `new` (0001–0004), `oldAA` (a second server of the old build, A/A control). Every system × 5 repeats, the arm order rotating with the repeat and the system; wall from request to reply; reply word and output SHA-256 compared across arms.
3. `summarize.py`, `make_report.sh`.

Binaries: `scripts/m4gb/build.sh n16_gf2 n20_gf2 n24_gf2 n28_gf2 n32_gf2` from the pinned commit, once with patches 0001–0003 and once with 0001–0004. Local container, one thread, not isolated; a stage diagnostic, no leaderboard row.

Pre-stated expectation and falsifier

Expectation: the paired ratio new/old falls with the number of unknowns, about 0.75× at 16 to 20 unknowns and about 0.6× at 24, near 0.9× on the degree-6 m = 3 systems (their time is less in `update()`), with the A/A ratio within a few percent of 1 on every cell and no disagreement. Falsifier: any disagreement in reply or output bytes; or a cell at 20 or more unknowns whose median paired ratio is not below the A/A band.

Results

Filled after the run.

Paired, per captured system: median wall of 5 requests per arm through `--serve`, one thread. `old` = patches 0001–0003, `new` = 0001–0004, `oldAA` = a second server of the `old` build (A/A noise control). Cell columns: median over the cell's systems; ratios are the median (min–max) over systems of the per-system ratio.

cell	vars	systems	`old` ms	`new` ms	`new`/`old`	systems `new` below	A/A `oldAA`/`old`
K-11-6-2	12	6	3.53	3.71	0.97 (0.87–1.23)	3/6	0.99 (0.96–1.03)
R-11-6-2	12	6	3.39	3.09	0.91 (0.87–1.24)	5/6	1.02 (0.98–1.07)
K-13-7-2	14	6	8.67	5.99	0.75 (0.63–0.94)	6/6	0.97 (0.78–1.01)
R-13-7-2	14	6	15.43	10.12	0.66 (0.48–1.40)	5/6	0.93 (0.72–1.09)
K-15-8-2	16	6	15.53	10.52	0.68 (0.65–0.89)	6/6	1.00 (0.99–1.03)
R-15-8-2	16	6	29.25	18.60	0.64 (0.45–0.81)	6/6	1.10 (0.74–1.45)
K-17-9-2	18	6	31.97	20.59	0.65 (0.57–0.79)	6/6	1.02 (1.00–1.25)
R-17-9-2	18	6	36.37	27.73	0.73 (0.60–0.94)	6/6	1.11 (0.76–1.24)

cell	vars	systems	old ms	new ms	new/old	systems new below	A/A oldAA/old
K-19-10-2	20	6	41.20	31.61	0.72 (0.68–0.79)	6/6	0.98 (0.85–1.11)
R-19-10-2	20	6	52.00	37.25	0.74 (0.40–0.83)	6/6	1.01 (0.98–1.14)
R-21-11-2	22	6	231.48	129.77	0.57 (0.51–0.69)	6/6	0.96 (0.88–1.14)
K-23-12-2	24	6	430.26	235.45	0.61 (0.51–0.67)	6/6	1.03 (0.95–1.10)
R-23-12-2	24	6	334.33	203.34	0.64 (0.56–0.79)	6/6	1.01 (0.89–1.04)
K-11-4-3	12	6	7000.49	6231.00	0.88 (0.87–0.93)	6/6	1.00 (0.96–1.01)
R-11-4-3	12	6	7199.09	6493.82	0.89 (0.88–0.92)	6/6	1.01 (0.95–1.03)

Disagreements between arms (reply word or output bytes): 0.